Provenance-Aware Agentic Knowledge Graphs for AI-Accelerated Scientific Discovery

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Focus Area: 11 – Achieving AI-Driven Autonomous Laboratories
Topic: C. AI-Accelerated Science: Correlation to Understanding

Project Summary / Abstract

The pace of scientific discovery is rapidly increasing, accelerated by recent advances in large language models (LLMs), generative artificial intelligence (AI) systems trained on vast amounts of data. These innovations are critical for societal progress and well-being, yet current experimental processes constrain research discovery. Traditional human-driven experimentation struggles to collect and analyze substantial amounts of data efficiently [], while current AI-driven tools lack the determinism and traceability needed for rigorous scientific workflows. For instance, AI systems can retrieve relevant documents and generate plausible text—however, they lack the ability to systematically compare new observations against a persistent representation of current scientific knowledge or autonomously generate new knowledge and hypotheses from integrated data with traceable provenance connecting back to the specific evidence that supports each inference. Without persistent knowledge infrastructure, every agent session starts from scratch—unable to build on prior findings, detect genuine anomalies against an accumulated model, or produce outputs that human researchers can audit and extend—resulting in a fragmented AI-driven discovery cycle.

To address these challenges, we will introduce **provenance-aware agentic knowledge graphs** (PA-AKG) as foundational infrastructure for AI-accelerated scientific discovery. A persistent research knowledge graph (KG) encodes the current state of scientific understanding—problems, datasets, methods, constraints, and evidence—linked by explicit semantic relations. This structured representation is the "current model" against which incoming scientific data is continuously compared. As new evidence emerges from literature or ongoing experiments, LLM-based agents compare new findings against KG-encoded expectations, detect divergences, and produce structured, evidence-grounded explanations with provenance chains. Confirmed hypotheses are written back to the KG, updating the current model and enabling the next discovery cycle. PA-AKG will offer a significant AI advantage over existing retrieval-augmented approaches. This closed-loop process will promote the AI-driven identification of scientific gaps, correlations in high-throughput data, and the generation of new testable hypotheses—accelerating scientific discovery and understanding.

The proposed work will design and implement the PA-AKG infrastructure, incorporating AI directly into experimental workflows for data collection, data analysis hypothesis generation, and data curation and sharing. We will evaluate PA-AKG against existing approaches, investigating the accuracy, traceability, and usability of our approach. This project will advance foundations for AI-accelerated autonomous research by designing techniques to persist scientific knowledge accumulation across research sessions, enabling AI to reason over an evolving model of current understanding and identify genuine gaps rather than re-derive the same correlations each session. This work will produce deployable tools, including an open-source implementation of the PA-AKG architecture, benchmarks, and empirical insights to lay the groundwork for a new generation of trustworthy AI-augmented science.

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Cover Sheet

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Primary Challenge:

11 – Achieving AI-Driven Autonomous Laboratories
C. AI-Accelerated Science: Correlation to Understanding

Senior/Key Personnel for Lead Institution and All Partner Institutions	
Institution Name	**Senior/Key Personnel Including the Director**
Virginia Tech	PI Brown; Co-PI Wang
Sandia National Labs	Co-PI Steinmetz

Summary budget information for lead institution and all partner institutions	
Institution	**Year 1 Budget $**
Virginia Tech (lead)	**[[TODO:]]**
Sandia National Labs	**[[TODO:]]**

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GRA: Jason Cusati, Computer Science, Virginia Tech

Project Narrative

Introduction The volume of scientific discovery and data are expanding at an unprecedented rate [26]. For example, the Department of Energy's (DOE) own research portfolio generates petabytes of experimental and simulation data annually [29] across facilities such as Brookhaven National Laboratory, Oak Ridge National Laboratory, and Sandia National Laboratories. Yet our capacity to synthesize, contextualize, and act on this accumulating knowledge has not kept pace with its production. This widening **knowledge integration gap** poses significant challenges for national research priorities, especially in computing and energy research, where the accumulation and generation of new knowledge is often fragmented and burdensome, leading to substantial duplication of effort and inefficient use of critical national laboratory resources.

Recent advances in large language models (LLMs) and generative artificial intelligence (AI) have produced tools of remarkable capability to support scientific discovery. LLM-based systems can retrieve and summarize vast bodies of literature [1], suggest experimental protocols [31], replicate evaluations [16], and generate plausible scientific text [10] across virtually any domain. The potential of LLM-based agents to accelerate DOE-relevant research is widely recognized and actively being pursued across the national laboratory system [23]. However, LLMs are prone to hallucinate—or generate plausible text without the ability to distinguish between well-supported scientific claims and speculative or incorrect assertions—across various research contexts including fake citations, incorrectly recalled experimental results, or duplicate hypothesis generation [15]. These outcomes can misdirect experimental programs and waste substantial resources [22]. Recently, agentic systems coordinating multiple AI agents [2, 9] and augmentation methods to retrieve knowledge from relevant documents [20, 32] aim to enhance research efforts through grounding and contextualizing model reasoning. However, fundamental challenges that constrain LLM utility for rigorous, reproducible scientific workflows remain:

Absence of persistent knowledge state. Current LLM-based systems lack the ability to recall prior sessions or the current state of research. This limitation prevents AI from building cumulative understanding, detecting contradictions from established previous findings, and avoiding redundant efforts. As a result, agents provide informative but not progressive insights, failing to learn from ongoing experiments.

Lack of traceable provenance. Scientific conclusions must be auditable. For instance, when an AI system suggests a hypothesis, a researcher must be able to trace that suggestion to specific experimental observations and theoretical arguments that support it. While current systems may surface source documents, they do not provide structured reasoning that connects evidence to conclusions. Without explicit provenance, hypotheses generated by AI cannot be rigorously evaluated or reliably used for further experimentation.

The lack of a structured and persistent representation of scientific knowledge limits AI agents' ability to effectively contribute to the scientific discovery process.

To address these challenges, this project proposes the design, implementation, and evaluation of **provenance-aware agentic knowledge graphs** (PA-AKG). PA-AKG is a foundational infrastructure for closed-loop, AI-integrated scientific discovery, addressing the challenges identified above by providing AI agents with a persistent, structured, and auditable representation of the current state of scientific understanding in a given domain. Knowledge graphs (KGs) have been proposed as a mechanism for representing complex, heterogeneous information through explicit semantic relationships [7, 11]. PA-AKG will encode scientific knowledge (*i.e.,* problems, datasets, experimental methods, theoretical frameworks, and empirical observations) as nodes representing existing knowledge, and define relationships between them as directed edges. As new evidence emerges, LLM-based agents will compare incoming observations against KG-encoded expectations. When divergences are detected, agents generate structured, evidence-grounded hypotheses, each carrying an explicit provenance chain that traces the hypothesis back to the specific nodes and edges in the knowledge graph that motivate it. Confirmed hypotheses are written back to the KG, updating the current model and enabling the next iteration of the discovery cycle.

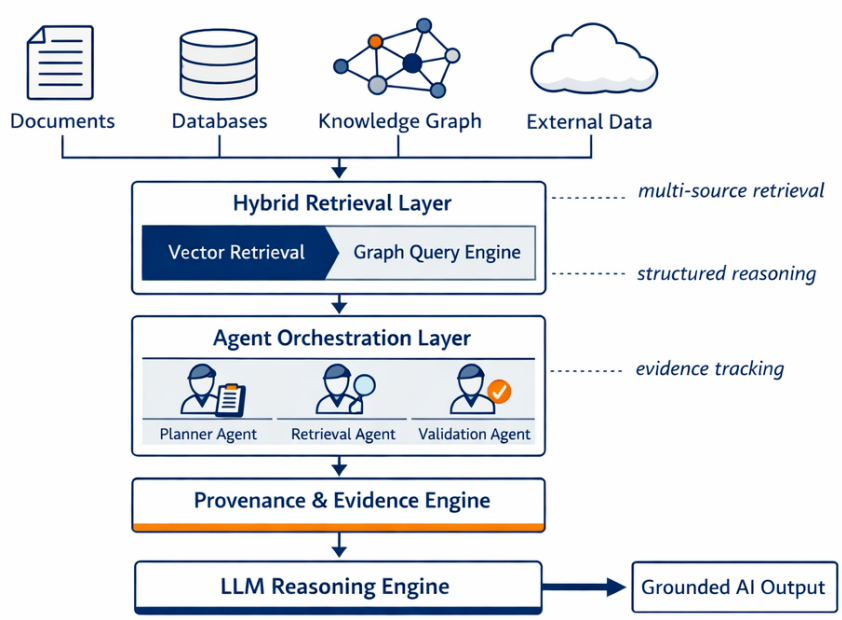

Figure 1: PA-AKG Architecture

Objectives. Our goal is to design and implement a provenance-first agentic KG architecture with three integrated layers: knowledge representation; automation and extraction; and agentic orchestration (see Figure 1). We will implement the PA-AKG architecture, developing infrastructure to automate the extraction and structuring of claim-evidence pairs with role-specific agents (*e.g.*, ranking agents, evaluation agents, and synthesis agents). The initial implementation will focus on research in the software engineering domain, following our prior work [6]. Software engineering, the processes and methods for developing and maintaining software systems, is also critical for scientific discovery and innovation [28], enabling infrastructure across domains including next-generation computing, national security, and biotechnology. We will conduct a comprehensive evaluation of our approach, comparing our system against RAG, GraphRAG [8], and static baseline approaches to compare persistence, traceability, and hypothesis quality. We will also conduct controlled user studies to obtain researcher perceptions of our system and assess its usability. Key risks include extraction pipeline reliability at corpus scale and KG schema generalization across scientific domains; these are mitigated by the staged evaluation design and secondary domain validation.

Research Plan. Our research plan involves three main phases:

Phase 1 (Months 1–3): Knowledge Representation Layer. We will first design the research KG schema with entity types, such as PROBLEM, DATASET, METHOD, CONSTRAINT, EVIDENCE—linked by semantic relations (SUPPORTS, CONTRADICTS, REQUIRES, etc.). We will conduct pilot studies (*e.g.*, surveys and interviews) to obtain insights from researchers to inform entity types and relationships for our initial implementation. Research problems will be modeled with identifiable assumptions, methods, and evaluation criteria, extracted from source documents with assumptions, constraints, datasets, metrics, and evidence spans anchored to source passages. A hybrid symbolic–semantic design pairs a property graph store with a vector index, supporting both structured relational queries and approximate similarity search. We will populate an initial KG from the software engineering research corpus motivated by prior work [6], validate entity extraction against manual ground-truth annotations, and establish baseline KG coverage metrics.

Phase 2 (Months 4–6): Automation, Extraction, and Agentic Orchestration. Next, we will implement the extraction and automation pipeline: structured prompting will produce claim–evidence pairs; span-level alignment will link each claim to character-level source passage spans; and canonicalization auditing will normalize, validate, and deduplicate entity references to canonical KG nodes. The agentic orchestration layer, adapting the Planning & Control pattern [19], introduces four role-specific agent types. *Ranking agents* score candidate research directions by tractability and evidence strength. *Continuation agents* will

propose follow-on analyses grounded in KG gaps. *Evaluation agents* will execute reproducible experimental workflows and write results back to the KG as structured evidence nodes. *Synthesis agents* will generate hypothesis artifacts—problem statement, supporting evidence chain, proposed test, predicted outcome—and curate them to the KG as HYPOTHESIS nodes with full provenance metadata. The provenance inheritance constraint will be enforced at every agent handoff, and each agent will receive a provenance context object and may only produce outputs that extend, not discard, the inherited evidence chain.

Phase 3 (Months 7–9): Evaluation and Generalization. We will conduct two comprehensive evaluations of our PA-AKG approach. The first will benchmark our system against existing approaches, including vector-based RAG, hybrid retrieval, and GraphRAG-style reasoning. We will construct a benchmark of human-annotated claim–evidence pairs from a research corpus. We will calculate span-level alignment, precision/recall, Cohen's κ with expert-labeled canonical entity clusters, and traceability completeness (*i.e.,* % of KG nodes with complete provenance chains). To enhance generalizability, a secondary scientific domain will be evaluated to assess cross-domain generalizability of the extraction pipeline and PA-AKG approach. Our second evaluation will assess the usability and utility of PA-AKG through controlled user studies with researchers in software engineering and secondary domains, collecting quantitative task-completion metrics and qualitative feedback on provenance chain utility.

Phase 4: Dissemination. We will pursue targeted dissemination of PA-AKG research findings to academic and practitioner audiences. Academic outputs will be submitted to premier venues in software engineering, AI, and machine learning, including the International Conference on Software Engineering (ICSE), the Conference on Empirical Software Engineering and Measurement (ESEM), and the AAAI Conference on Artificial Intelligence. We will submit to tool, demo, and tutorial tracks at these venues to provide hands-on exposure of PA-AKG to active research communities, lowering the barrier to adoption across research programs. To reach the research practitioner community, we will release the PA-AKG implementation as an open-source package with comprehensive documentation, worked examples, and integration guides targeting national laboratory researchers and the broader scientific computing community. We will present findings at Sandia National Laboratories and partner institutions to gather practitioner feedback, surface real-world deployment barriers, and assess adoption pathways within DOE workflows. This phase will conclude with an impact assessment evaluating how PA-AKG changes researchers' ability to identify genuine knowledge gaps and generate traceable hypotheses—informing the design of future AI-accelerated scientific discovery systems and contributing new foundations for trustworthy AI in DOE's research mission.

Research Team. This project aligns with the prior work and trajectory of the research team. **PI Brown** and GRA Cusati explored how LLMs propagate research-based evidence [4] and outlined researchers' desire for structured knowledge accumulation [6] in software engineering contexts. For this work, Dr. Brown will lead the overall execution of the project—including the PA-AKG implementation and research plan. **Co-PI Wang** investigates LLMs in the context of science and society, with substantial experience studying the applications of LLMs (*i.e.,* [3, 25]) and agents (*i.e.,* [17, 13]) in scientific settings and earning an NSF CAREER award to extract patterns from research data [30]. Dr. Wang will inform the design of the PA-AKG architecture, particularly focusing on knowledge retrieval and extraction, multi-agent orchestration, and knowledge persistence, and the benchmark evaluation [24]. **Co-PI Steinmetz** is a researcher at Sandia National Labs with expertise in human-machine teaming [5, 21], generative AI [12, 14], and trust in AI systems [27, 18]. Dr. Steinmetz will inform the PA-AKG design and support our research efforts to evaluate the usability and trustworthiness of our approach in practical research contexts. **GRA Cusati** is a Ph.D. student in the Department of Computer Science at Virginia Tech. He has extensive industry experience implementing complex production software in various roles. Jason will lead the design and development of the PA-AKG infrastructure and the overall data collection and analysis efforts.

Resources. Virginia Tech's Advanced Research Computing [] provides GPU cluster infrastructure for large-scale KG construction and LLM inference workloads. The team will access publicly available scientific literature corpora (*e.g.,* ACM Digital Library, arXiv, IEEE Xplore) and pre-trained language models (*e.g.,* Llama, GPT-4, etc.) for extraction and orchestration. Sandia National Laboratories will contribute access to DOE-relevant scientific corpora for secondary domain validation, ensuring the research addresses realistic national laboratory knowledge integration challenges.

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